

The Most Promising BiCRS Pathways in the United Kingdom

Biomass Carbon Removal and Storage (BiCRS) converts waste and residual biomass into durable CO₂ removal — either by capturing and geologically storing the biogenic CO₂ released when biomass is used for energy or fuel, or by locking the carbon away directly. This brief summarises where the UK's most promising opportunities lie, drawn from the interactive **UK BiCRS Atlas** (<https://hausfath.github.io/bicrs-map-uk/>), which resolves feedstock, storage, and best-use recommendations to each of the UK's 41 NUTS-2 / ITL-2 regions.

The Atlas ranks pathways using Frontier's procurement KPIs — **carbon-removal efficiency first, then emissions-avoiding co-products, then wider co-benefits** — screened against feedstock moisture and density, distance and cost to reach CO₂ storage, soil-nutrient status, and whether a facility suitable for retrofit already exists nearby. Purpose-grown energy crops are excluded by design; the focus is genuinely residual and waste biomass.

Headline: across residual UK biomass, the best-use recommendation for each region sums to roughly **12.8 Mt CO₂ /yr of durable removal** — a screening estimate of what a mature, right-sized UK BiCRS sector could deliver from wastes and residues alone, before any dedicated energy crops. The UK is unusually well positioned: it pairs a modest but real residue base with the largest offshore CO₂ storage resource in Europe and two storage hubs already in construction.

1. Feedstock availability

The UK is not a biomass-rich country by land area, but its **waste streams are substantial and highly concentrated near population and industry** — favourable for BiCRS, which rewards feedstock that is already aggregated. Recoverable UK totals (Mt/yr, oven-dry except MSW):

Feedstock	Recoverable supply	Character & best fit
Municipal solid waste (total)	~25.7 Mt/yr (~55% biogenic)	Concentrated in cities; already collected → WtE + CCS
Animal manure	~12.0 Mt/yr	Wet, distributed across the livestock west/north → AD + CCS or HTL bio-oil
Agricultural residues	~6.3 Mt/yr	Straw/stover, concentrated in the arable east → BECCS / injection
Human / WWTP biosolids	~1.0 Mt/yr	Wet, at large sewage works → co-digestion / injection
Forestry residues	~0.5 Mt/yr	Small; the UK is import-dependent for wood

Two features stand out. First, **MSW and manure dominate the tonnage** — both wet or heterogeneous streams that suit anaerobic digestion, waste-to-energy, or hydrothermal routes rather than dry combustion. Second, the geography is clean: **arable dry residues cluster in East Anglia and the east**, while **wet manure dominates the livestock west and north** (Wales, South West England, Scotland, Northern Ireland). That split largely determines the recommended pathway region-by-region.

Manure and crop-residue supply are distributed using the **UK June Agricultural Census** (housed cattle/pig/poultry numbers for manure; arable-land area for straw), so recoverable manure tracks where livestock actually are — near-zero in London and the conurbations, concentrated in Northern Ireland, West Wales, the South West, North Yorkshire and southern Scotland.

2. Retrofit opportunities

The UK's strongest near-term advantage is a **large existing biomass and waste-combustion fleet that can be retrofitted with capture**, avoiding greenfield build. The Atlas maps 26 candidate biogenic point sources:

- **Biomass power (11 sites, ~18.9 Mt biogenic CO₂/yr of throughput).** Dominated by **Drax** (~12 Mt/yr biogenic CO₂ — the single largest BECCS opportunity in Europe), plus **Lynemouth, MGT Teesside, Wilton 10 / Sembcorp Teesside, Margam (Port Talbot)**, and a cluster of ~40 MW plants (Snetterton, Sleaford, Brigg, Kent/Sandwich, Ince/Mersey). These anchor **BECCS** in East Anglia and the Humber/Teesside and Mersey clusters.
- **Waste-to-energy (13 sites, ~3.0 Mt biogenic CO₂/yr).** The urban EfW fleet — Runcorn, Cory Riverside/Belvedere, Edmonton, Lakeside, Earls Gate (Scotland) — anchors **WtE + CCS** across the English cities and central belt.
- **Anaerobic digestion (existing UK biogas cluster).** Hundreds of farm and food-waste digesters already operate; retrofitting biogas-upgrading CO₂ vents is among the cheapest capture available and anchors **AD + CCS** across the livestock regions.

Crucially, these anchors sit close to the **CO₂ storage hubs now in construction**: the **Northern Endurance Partnership / East Coast Cluster** (Teesside–Humber) and **Liverpool Bay / HyNet** (Mersey), with **Acorn** (Scotland) and **Viking** (Humber) planned. In the Atlas, **36 of 41 UK regions score "good" storage access** — a far higher share than most of Europe — because these offshore stores are within economic transport reach. Modelled delivered CO₂-transport cost is low (median ~50/tCO₂, range 13–73), reflecting short routes to the North Sea.

3. Total CO₂ removal potential by pathway

Applying the best-use recommendation to each region gives the following screening-level CDR potential. Pathways are listed by total potential; "regions" is how many of the 41 pick that pathway as their best use.

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost	Where
WtE + CCS (capture on energy-from-waste)	18	4.1 Mt CO₂/yr	~55%	~\$100–200/t	Urban England (incl. London), central-belt Scotland
HTL bio-oil (hydrothermal liquefaction of wet residues)	8	3.8 Mt CO₂/yr	~50%	\$200–400/t	Livestock west/north far from anchors: N. Ireland, Highlands, SW England

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost	Where
AD + CCS (biogas upgrading capture)	9	2.3 Mt CO ₂ /yr	~37%	\$145–300/t	Livestock regions with digesters: Wales, Yorkshire, Shropshire, E. Midlands
BECCS (biomass power + capture)	2	1.6 Mt CO ₂ /yr	~80%	\$200–225/t	East Anglia, Essex (near biomass plants + arable straw)
Biomass waste injection	4	1.0 Mt CO ₂ /yr	~90%	\$125–285/t	Dry-residue regions right on top of storage: Hampshire, Teesside, NE Scotland
Total	41	~12.8 Mt CO₂/yr			

How to read this. The ranking reflects *where each pathway is the best local fit*, not the ceiling for any one technology. Two points for prioritisation:

- **Biggest and lowest-regret: WtE + CCS.** It is now the single largest pathway by both region count and potential (~4.1 Mt/yr across 18 regions), because MSW is the UK's largest biogenic waste stream and is concentrated in cities that already run energy-from-waste plants. It removes carbon *and* decarbonises waste the UK must manage anyway, converts a high share of feedstock carbon (~55%), and retrofits an existing urban fleet. **BECCS** and **injection** are the highest-efficiency routes (80–90%); **Drax alone** (BECCS) represents removal of the same order as the entire recommended-pathway total if fully captured — the single most consequential UK decision.
- **Wet-residue routes for the livestock country: HTL bio-oil and AD + CCS.** These serve the distributed manure of the west and north, where combustion is a poor fit. AD + CCS is cheap where digesters exist; HTL bio-oil is the fallback for wet residues too far from a capture anchor or store, since the densified oil can be injected without gaseous-CO₂ transport.

Bottom line for the UK

The UK's BiCRS advantage is **not feedstock volume — it is the combination of an existing combustion/AD fleet to retrofit and the best offshore CO₂ storage access in Europe**. The fastest, highest-quality tonnes come from **capturing CO₂ on the biomass-power and energy-from-waste plants already clustered around the Teesside/Humber, Mersey, and central-belt storage hubs**. Wet-residue routes (AD + CCS, HTL bio-oil) then extend BiCRS into the livestock regions that combustion cannot serve. Explore any region's rationale, ranked alternatives, and delivered costs in the interactive Atlas.

Source: UK BiCRS Atlas (<https://hausfath.github.io/bicrs-map-uk/>), the UK subset of Frontier's BiCRS Atlas. Manure and crop residues distributed by the UK June Agricultural Census (Eurostat Farm Structure Survey); MSW and biosolids by population; forestry from JRC ENSPRESO — all scaled to UK country totals. Storage from JRC CO₂StoP and curated UK CCS projects; facilities curated with capacity-estimated biogenic CO₂. All figures are screening-level estimates carrying material uncertainty and are not a substitute for project-level diligence. Grounded in Frontier's BiCRS purchasing point of view.